



Top 30 CAT Averages Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by $2x$ kg. If the weight of E is 12 kg more than that of D, then the value of x is

- A 2
- B 0.5
- C 1
- D 1.5

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There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by $2x$ kg. If the weight of E is 12 kg more than that of D, then the value of x is

A) 2
C) 1

B) 0.5

$$\begin{aligned} A+B+C &= 3P \\ A+B+C+D &= 4(P-x) = 4P-4x \\ \hline D &= P-4x \end{aligned}$$

P
 $P-x$

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Question 2

In a company, 20% of the employees work in the manufacturing department. If the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company, then the ratio of the average salary obtained by the manufacturing employees to the average salary obtained by the nonmanufacturing employees is

- A 6:5
- B 4:5
- C 5:4
- D 5:6

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In a company, 20% of the employees work in the manufacturing department. If the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company, then the ratio of the average salary obtained by the manufacturing employees to the average salary obtained by the nonmanufacturing employees is

A) 6:5
B) 4:5
C) 5:4

Handwritten notes: Total = 100, Manu = 20, Non Manu = 80, 20M1, 4:5, 20M1

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Question 3

The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is

- A 1 : 4
- B 1 : 2
- C 4 : 1
- D 3 : 1

cracku

The average weight of students in a class increases by 600 gm when some new students join the class. If the average weight of the new students is 3 kg more than the average weight of the original students, then the ratio of the number of original students to the number of new students is

A) 1:4
B) 1:2
C) 4:1
D) 3:1

Handwritten notes: w , $w+2$, $w+0.6$, A , B , $A+B$

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Question 4

Manu earns ₹4000 per month and wants to save an average of ₹550 per month in a year. In the first nine months, his monthly expense was ₹3500, and he foresees that, tenth month onward, his monthly expense will increase to ₹3700. In order to meet his yearly savings target, his monthly earnings, in rupees, from the tenth month onward should be

- A 4400

- B 4200
- C 4300
- D 4350

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A) 4400
B) 4200
C) 4300

Monthly Savings $\rightarrow 4000 - 3500 = 500$
 $500 \times 9 + S$

VIDEO SOLUTION

Question 5

In an examination, the average marks of students in sections A and B are 32 and 60, respectively. The number of students in section A is 10 less than that in section B. If the average marks of all the students across both the sections combined is an integer, then the difference between the maximum and minimum possible number of students in section A is

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Answer: _____

$$\frac{A}{A}$$

$$\frac{32}{(32)}$$

$$\frac{46A + 230 + 5}{A + 5}$$

$$\frac{A+10}{B}$$

$$\frac{60}{(60)}$$

$$\frac{32A + 60A + 600}{2A + 10}$$

$$\frac{92A + 600}{2A + 10}$$

$$\frac{46A + 300}{A + 5}$$

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Question 6

The average of three integers is 13. When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is

- A 3
- B 4
- C 5
- D 1

The average of three integers is 13. When a natural number n is included, the average of these four integers remains an odd integer. The minimum possible value of n is _____

A) 3
B) 4
C) 5
D) 1

$13 \times 3 = 39$

$\frac{39+n}{4} = \text{odd integer}$



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Question 7

If a and b are non-negative real numbers such that $a + 2b = 6$, then the average of the maximum and minimum possible values of $(a + b)$ is

- A 3
- B 4
- C 3.5
- D 4.5

If a and b are non-negative real numbers such that $a + 2b = 6$, then the average of the maximum and minimum possible values of $(a + b)$ is _____

A) 3
B) 4
C) 3.5
D) 4.5

$a + b = 6 - b$

$(a+b)_{\max} = 6$
 $b_{\min} = 0$



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Question 8

A batsman played $n + 2$ innings and got out on all occasions. His average score in these $n + 2$ innings was 29 runs and he scored 38 and 15 runs in the last two innings. The batsman scored less than 38 runs in each of the first n innings. In these n innings, his average score was 30 runs and lowest score was x runs. The smallest possible value of x is

- A 4

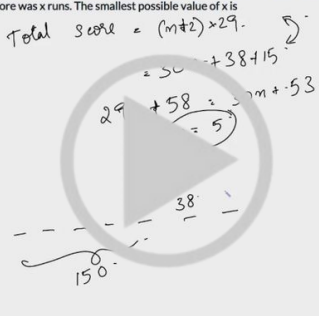
B 3

C 2

D 1

A batsman played $n+2$ innings and got out on all occasions. His average score in these $n+2$ innings was 29 runs and he scored 38 and 15 runs in the last two innings. The batsman scored less than 38 runs in each of the first n innings. In these n innings, his average score was 30 runs and lowest score was x runs. The smallest possible value of x is

Handwritten solution:

$$\text{Total score} = (n+2) \times 29$$
$$= 50 + 38 + 15$$
$$29 + 58 = n + 53$$
$$n = 15$$
$$150$$


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Question 9

In a football tournament, a player has played a certain number of matches and 10 more matches are to be played. If he scores a total of one goal over the next 10 matches, his overall average will be 0.15 goals per match. On the other hand, if he scores a total of two goals over the next 10 matches, his overall average will be 0.2 goals per match. The number of matches he has played is

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Answer: _____

Handwritten solution:

$$\frac{G+1}{N+10} = 0.15$$
$$\frac{G+2}{N+10} = 0.2$$
$$10$$
$$1 \quad 2$$


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Question 10

The arithmetic mean of scores of 25 students in an examination is 50. Five of these students top the examination with the same score. If the scores of the other students are distinct integers with the lowest being 30, then the maximum possible score of the toppers is

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Answer: _____

Handwritten notes on a whiteboard:

- 25
- 50
- 1250
- 1250 - 5T
- 30 + 31 ... 49
- 5T



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Question 11

Onion is sold for 5 consecutive months at the rate of Rs 10, 20, 25, 25, and 50 per kg, respectively. A family spends a fixed amount of money on onion for each of the first three months, and then spends half that amount on onion for each of the next two months. The average expense for onion, in rupees per kg, for the family over these 5 months is closest to

- A 26
- B 18
- C 16
- D 20

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Handwritten notes on a whiteboard:

- Per Kg
- Rs-10
- Rs-20
- Rs-25
- Rs-25
- Rs-50
- 100F
- 200F
- 50F
- 50F
- 50F
- Amu



VIDEO SOLUTION

Question 12

Five students, including Amit, appear for an examination in which possible marks are integers between 0 and 50, both inclusive. The average marks for all the students is 38 and exactly three students got more than 32. If no two students got the same marks and Amit got the least marks among the five students, then the difference between the highest and lowest possible marks of Amit is

- A 22
- B 21
- C 24

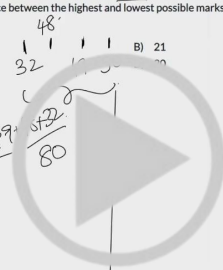
D 20

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A) 22
B) 21
C) 24
D) 20

Handwritten notes: 48 , 32 , 10 , 30 , 20 , $50 + 49 + \dots + 22$, 80 , $\text{Total Marks} = 38 \times 5 = 190$, min for Amit



VIDEO SOLUTION



CAT Syllabus PDF



Question 13

The average of 30 integers is 5. Among these 30 integers, there are exactly 20 which do not exceed 5. What is the highest possible value of the average of these 20 integers?

- A 3.5
- B 5
- C 4.5
- D 4

cracku

The average of 30 integers is 5. Among these 30 integers, there are exactly 20 which do not exceed 5. What is the highest possible value of the average of these 20 integers?

Handwritten notes: $\text{Average of 30 integers} = 5$, $\text{Total} = 30 \times 5 = 150$, $20 \leq 5$, $10 > 5$, $10 \geq 6$



VIDEO SOLUTION

Question 14

The average marks of a student in 10 papers are 80. If the highest and the lowest scores are not considered, the average is 81. If his highest score is 92, find the lowest.

- A 55

- B 60
- C 62
- D Cannot be determined

The average marks of a student in 10 papers are 80. If the highest and the lowest scores are not considered, the average is 81. If his highest score is 92, find the lowest.

A) 55
B) 60
C) 62
D) Cannot be determined

$Total = 80 \times 10 = 800$

H, L, 8 others

81

VIDEO SOLUTION

Question 15

Total expenses of a boarding house are partly fixed and partly varying linearly with the number of boarders. The average expense per boarder is Rs. 700 when there are 25 boarders and Rs. 600 when there are 50 boarders. What is the average expense per boarder when there are 100 boarders?

- A 550
- B 580
- C 540
- D 560

Total expenses of a boarding house are partly fixed and partly varying linearly with the number of boarders. The average expense per boarder is Rs. 700 when there are 25 boarders and Rs. 600 when there are 50 boarders. What is the average expense per boarder when there are 100 boarders?

A) 550
B) 580
C) 540
D) 560

$TC = F + mV$

$17500 = F + 25V$

$30000 = F + 50V$

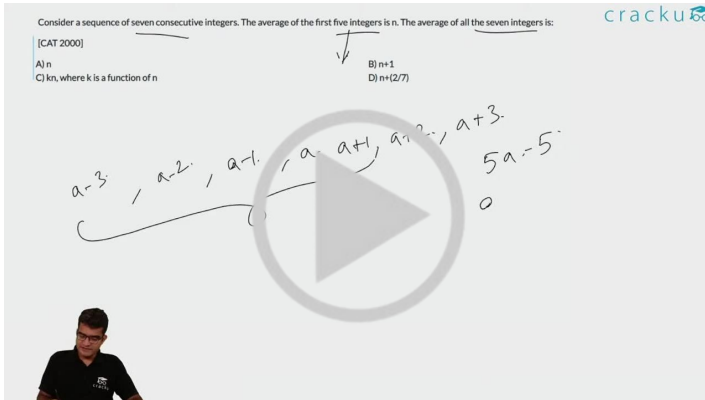
VIDEO SOLUTION

 **CAT Formula PDF** 

Question 16

Consider a sequence of seven consecutive integers. The average of the first five integers is n . The average of all the seven integers is:

- A n
- B $n+1$
- C kn , where k is a function of n
- D $n+(2/7)$



 VIDEO SOLUTION

Question 17

Three classes X, Y and Z take an algebra test.

The average score in class X is 83.

The average score in class Y is 76.

The average score in class Z is 85.

The average score of all students in classes X and Y together is 79.

The average score of all students in classes Y and Z together is 81.

What is the average for all the three classes?

- A 81
- B 81.5
- C 82
- D 84.5

Three classes X, Y and Z take an algebra test.
 The average score in class X is 83.
 The average score in class Y is 76.
 The average score in class Z is 85.
 The average score of all students in classes X and Y together is 79.
 The average score of all students in classes Y and Z together is 81.
 What is the average for all the three classes?

A) 81
 B) 82

VIDEO SOLUTION

Question 18

Amol was asked to calculate the arithmetic mean of 10 positive integers, each of which had 2 digits. By mistake, he interchanged the 2 digits, say a and b , in one of these 10 integers. As a result, his answer for the arithmetic mean was 1.8 more than what it should have been. Then $|b - a|$ equals

- A 1
- B 2
- C 3
- D None of these

Amol was asked to calculate the arithmetic mean of 10 positive integers, each of which had 2 digits. By mistake, he interchanged the 2 digits, say a and b , in one of these 10 integers. As a result, his answer for the arithmetic mean was 1.8 more than what it should have been. Then $|b - a|$ equals

A) 1
 B) 2
 C) 3
 D) None of these

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Question 19

Ten years ago, the ages of the members of a joint family of eight people added up to 231 years. Three years later, one member died at the age of 60 years and a child was born during the same year. After another three years, one more member died, again at 60, and a child was born during the same year. The current average age of this eight-member joint family is nearest to

- A 23 years

- B 22 years
- C 21 years
- D 25 years
- E 24 years

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[CAT 2007]

A) 23 years
B) 22 years
C) 21 years
D) 25 years
E) 24 years

Handwritten notes: 10 yrs, 231 years, 55 yr, 95 yr, 22 years, 25 years, 8, 2.

VIDEO SOLUTION

Question 20

Consider the set $S = \{2, 3, 4, \dots, 2n+1\}$, where n is a positive integer larger than 2007. Define X as the average of the odd integers in S and Y as the average of the even integers in S . What is the value of $X - Y$?

- A 0
- B 1
- C $(1/2)*n$
- D $(n+1)/2n$
- E 2008

Consider the set $S = \{2, 3, 4, \dots, 2n+1\}$, where n is a positive integer larger than 2007. Define X as the average of the odd integers in S and Y as the average of the even integers in S . What is the value of $X - Y$?

A) 0
B) 1
C) $(1/2)*n$
D) $(n+1)/2n$
E) 2008

Handwritten notes: Odd Integers = $\{3, 5, 7, \dots, 2n+1\}$, Even Integers = $\{2, 4, 6, 8, \dots, 2n\}$, Arithmetic Progression.

VIDEO SOLUTION

Question 21

A class consists of 20 boys and 30 girls. In the mid-semester examination, the average score of the girls was 5 higher than that of the boys. In the final exam, however, the average score of the girls dropped by 3 while the average score of the entire class increased by 2. The increase in the average score of the boys is

A class consists of 20 boys and 30 girls. In the mid-semester examination, the average score of the girls was 5 higher than that of the boys. In the final exam, however, the average score of the girls dropped by 3 while the average score of the entire class increased by 2. The increase in the average score of the boys is _____.

Handwritten solution:

Let the average score of boys in the mid-semester exam be A and the average score of girls be G .

Given: $G = A + 5$

In the final exam, the average score of girls dropped by 3, so it became $G - 3 = A + 5 - 3 = A + 2$.

The average score of the entire class increased by 2. Let the new average score of the entire class be $A + 2$.

Let the increase in the average score of the boys be x . Then the new average score of boys is $A + x$.

Using the formula for the average score of the entire class:

$$\frac{\text{Total score of boys} + \text{Total score of girls}}{\text{Total number of students}} = \text{Average score of the entire class}$$
$$\frac{20(A + x) + 30(A + 2)}{20 + 30} = A + 2$$
$$\frac{20A + 20x + 30A + 60}{50} = A + 2$$
$$\frac{50A + 20x + 60}{50} = A + 2$$
$$50A + 20x + 60 = 50(A + 2)$$
$$50A + 20x + 60 = 50A + 100$$
$$20x = 100 - 60$$
$$20x = 40$$
$$x = 2$$

The increase in the average score of the boys is 2.

 VIDEO SOLUTION

CAT Percentile Predictor



Question 22

The average height of 22 toddlers increases by 2 inches when two of them leave this group. If the average height of these two toddlers is one-third the average height of the original 22, then the average height, in inches, of the remaining 20 toddlers is

The average height of 22 toddlers increases by 2 inches when two of them leave this group. If the average height of these two toddlers is one-third the average height of the original 22, then the average height, in inches, of the remaining 20 toddlers is

Diagram illustrating the problem:

- Original group: 22 toddlers, average height $A + 2$.
- Two toddlers leave: average height $A/3$.
- Remaining group: 20 toddlers, average height A .

Equation derived from the diagram:

$$A + 2 = \frac{20A + 2(A/3)}{22}$$

 VIDEO SOLUTION

Question 23

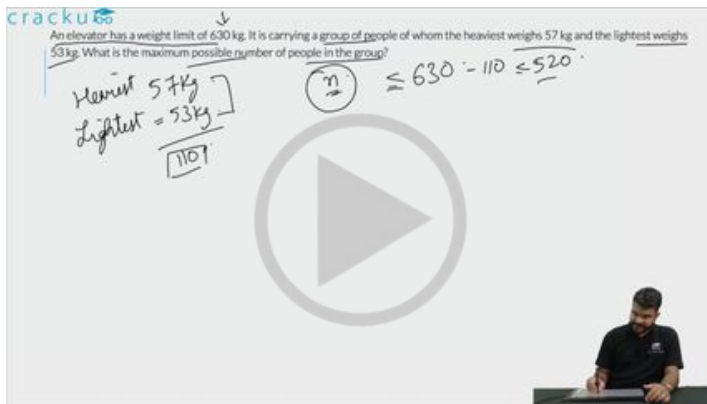
An elevator has a weight limit of 630 kg. It is carrying a group of people of whom the heaviest weighs 57 kg and the lightest weighs 53 kg. What is the maximum possible number of people in the group?

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An elevator has a weight limit of 630 kg. It is carrying a group of people of whom the heaviest weighs 57 kg and the lightest weighs 53 kg. What is the maximum possible number of people in the group?

Heaviest = 57 kg
Lightest = 53 kg
110

$n \leq 630 - 110 \leq 520$



VIDEO SOLUTION

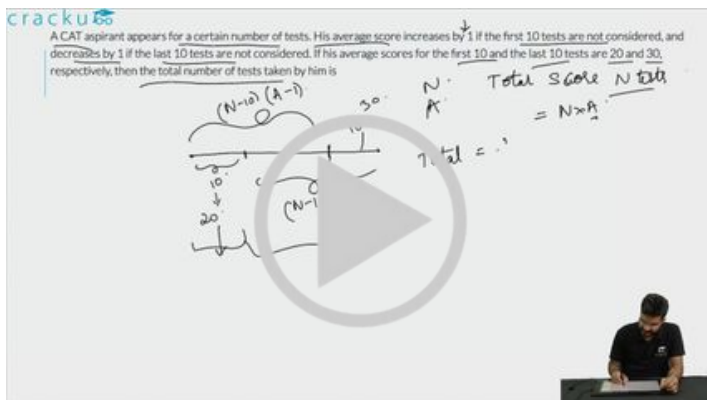
Question 24

A CAT aspirant appears for a certain number of tests. His average score increases by 1 if the first 10 tests are not considered, and decreases by 1 if the last 10 tests are not considered. If his average scores for the first 10 and the last 10 tests are 20 and 30, respectively, then the total number of tests taken by him is

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$(N-10)(A-1)$
 $(N-10)$
 20
 $(N-1)$
 30
 N
 A
 $Total\ Score = N \times A$



VIDEO SOLUTION

cracku

CAT Score Calculator



Question 25

In an apartment complex, the number of people aged 51 years and above is 30 and there are at most 39 people whose ages are below 51 years. The average age of all the people in the apartment complex is 38 years. What is the largest possible average age, in years, of the people whose ages are below 51 years?

- A 27
- B 25
- C 26

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In an apartment complex, the number of people aged 51 years and above is 30 and there are at most 39 people whose ages are below 51 years. The average age of all the people in the apartment complex is 38 years. What is the largest possible average age, in years, of the people whose ages are below 51 years?

$P \geq 51 \Rightarrow 30$
 $Q < 51 \Rightarrow 39 \cdot x$
 $\frac{30 + 39x}{69} = 38$
 $30 + 39x = 2622$
 $39x = 2592$
 $x = 66.46$
 $x \leq 39$
 $x = 39$

Total

VIDEO SOLUTION

Question 26

Ramesh and Gautam are among 22 students who write an examination. Ramesh scores 82.5. The average score of the 21 students other than Gautam is 62. The average score of all the 22 students is one more than the average score of the 21 students other than Ramesh. The score of Gautam is

- A 53
- B 51
- C 48
- D 49

cracku

Ramesh and Gautam are among 22 students who write an examination. Ramesh scores 82.5. The average score of the 21 students other than Gautam is 62. The average score of all the 22 students is one more than the average score of the 21 students other than Ramesh. The score of Gautam is

82.5
 62
 $22 \times (a+1) = 21 \times a$
 $22a + 22 = 21a$
 $a = -22$

VIDEO SOLUTION

Question 27

There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

There are four numbers such that average of first two numbers is 1 more than the first number, average of first three numbers is 2 more than average of first two numbers, and average of first four numbers is 3 more than average of first three numbers. Then, the difference between the largest and the smallest numbers, is

Answer: _____

Handwritten notes:

$$\begin{aligned} \frac{A+B}{2} &= A+1 \\ A+B &= 2A+2 \\ B &= A+2 \end{aligned}$$

Diagram showing numbers A, B, C, D with arrows indicating relationships: A to B is A+2, A to C is A+3, and A to D is A+4.



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Question 28

The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is

The average of three distinct real numbers is 28. If the smallest number is increased by 7 and the largest number is reduced by 10, the order of the numbers remains unchanged, and the new arithmetic mean becomes 2 more than the middle number, while the difference between the largest and the smallest numbers becomes 64. Then, the largest number in the original set of three numbers is

Answer: _____

Handwritten notes:

$$\begin{aligned} a < b < c \\ a+7 < b < c-10 \\ \frac{a+7+b+c-10}{3} &= b+2 \\ \frac{a+b+c-3}{3} &= 3b \\ a+b+c &= 84 \end{aligned}$$

Equation: $a+b+c = 84 \quad \text{--- (1)}$

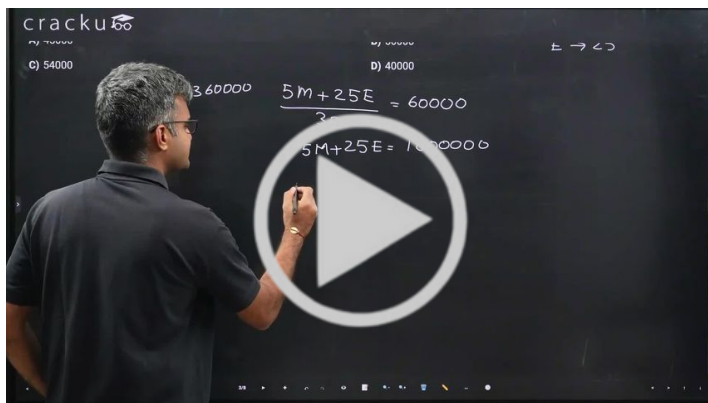


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Question 29

The average salary of 5 managers and 25 engineers in a company is 60000 rupees. If each of the managers received 20% salary increase while the salary of the engineers remained unchanged, the average salary of all 30 employees would have increased by 5%. The average salary, in rupees, of the engineers is

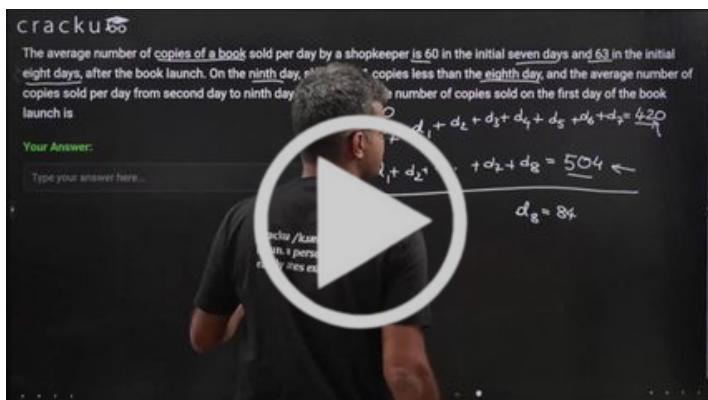
- A 45000
- B 50000
- C 54000
- D 40000



VIDEO SOLUTION

Question 30

The average number of copies of a book sold per day by a shopkeeper is 60 in the initial seven days and 63 in the initial eight days, after the book launch. On the ninth day, she sells 11 copies less than the eighth day, and the average number of copies sold per day from second day to ninth day becomes 66. The number of copies sold on the first day of the book launch is



VIDEO SOLUTION



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Answers

1.C	2.B	3.C	4.A	5.63	6.C	7.D	8.C
9.10	10.92	11.B	12.D	13.C	14.B	15.A	16.B
17.B	18.B	19.E	20.B	21.A	22.C	23.11	24.60
25.D	26.B	27.15	28.70	29.C	30.49		

Explanations

1. C

Let us assume that A, B, C, D, and E weights are a, b, c, d, and e.

1st condition

$$\frac{(a+b+c)}{3} - \frac{(a+b+c+d)}{4} = x$$

2nd condition

$$\frac{(a+b+c+e)}{4} - \frac{(a+b+c)}{3} = 2x$$

Adding both the equations, we get:

$$\frac{(e-d)}{4} = 3x$$

$$\Rightarrow \frac{(e-d)}{4} = 3x \Rightarrow e - d = 12x$$

Given that $12x = 12 \Rightarrow x = 1$.

 VIDEO SOLUTION

2. B

Let the number of total employees in the company be $100x$, and the total salary of all the employees be $100y$.

It is given that 20% of the employees work in the manufacturing department, and the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company.

Hence, the total number of employees in the manufacturing department is $20x$, and the total salary received by them is $(100y/6)$

Average salary in the manufacturing department = $(100y/6 \cdot 20x) = 5y/6x$

Similarly, the total number of employees in the nonmanufacturing department is $80x$, and the total salary received by them is $(500y/6)$

Hence, the average salary in the nonmanufacturing department = $(500y/6 \cdot 80x) = 25y/24x$

Hence, the ratio is:- $(5y/6x) : (25y/24x)$

$$\Rightarrow 120 : 150 = 4 : 5$$

The correct option is B

 VIDEO SOLUTION

3. C

Let the original number of students be 'n' whose average weight is 'x'

Let the number of students added be 'm' and the average weight will be $x + 3$

We need to find the value of $n : m$

It is given, average weight of students in a class increased by 0.6 after new students are added.

Therefore,

$$\frac{nx+m(x+3)}{n+m} = x + 0.6$$

$$nx + mx + 3m = mx + nx + 0.6n + 0.6m$$

$$2.4m = 0.6n$$

$$4m = n$$

$$\frac{n}{m} = \frac{4}{1}$$

The answer is option C.

 VIDEO SOLUTION

4. A

Savings target in a year = 550×12 = Rs 6600

Saving in first 9 months = $9(4000-3500)$ = Rs 4500

Saving for remaining 3 months should be 6600-4500, i.e. Rs 2100

Savings for each month in last 3 months = $\frac{2100}{3}$ = Rs 700

It is given, monthly expenses in last 3 months = Rs 3700

This implies, his monthly earnings from 10th month should be 3700+700, i.e. Rs 4400

The answer is option A.

 VIDEO SOLUTION

5. 63

Let the number of students in section A and B be a and b, respectively.

It is given, $a = b - 10$

$\frac{32a+60b}{a+b}$ is an integer

$$\frac{32a+60(a+10)}{a+a+10} = k$$

$$\frac{46a+300}{a+5} = k$$

$$k = \frac{46(a+5)}{a+5} + \frac{70}{a+5}$$

$$k = 46 + \frac{70}{a+5}$$

a can take values 2, 5, 9, 30, 65

Difference = $65 - 2 = 63$

 VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile



6. C

It is given that average of three numbers is 13.

Sum = $3 \times 13 = 39$

It is given, $\frac{39+n}{4}$ is a odd number.

Minimum value $\frac{39+n}{4}$ can take such that n is a natural number is 11

$$\frac{39+n}{4} = 11$$

$$n = 5$$

The answer is option C.

 VIDEO SOLUTION

7. **D**

$$a + 2b = 6$$

From the above equation, we can say that maximum value b can take is 3 and minimum value b can take is 0.

$$a + b + b = 6$$

$$a + b = 6 - b$$

a + b is maximum when b is minimum, i.e. b = 0

$$\text{Maximum value of } a + b = 6 - 0 = 6$$

a + b is minimum when b is maximum, i.e. b = 3

$$\text{Minimum value of } a + b = 6 - 3 = 3$$

$$\text{Average} = \frac{6+3}{2} = 4.5$$

The answer is option D.

 VIDEO SOLUTION

8. **C**

$$\text{Given, } \frac{\text{sum of scores in } n \text{ matches} + 38 + 15}{n+2} = 29$$

$$\text{Given, } \frac{\text{sum of scores in } n \text{ matches}}{n} = 30$$

$$\Rightarrow 30n + 53 = 29(n+2) \Rightarrow n=5$$

$$\text{Sum of the scores in 5 matches} = 29 \times 7 - 38 - 15 = 150$$

Since the batsmen scored less than 38, in each of the first 5 innings. The value of x will be minimum when remaining four values are highest

$$\Rightarrow 37 + 37 + 37 + 37 + x = 150$$

$$\Rightarrow x = 2$$

 VIDEO SOLUTION

9. **10**

Let Total matches played be n and in initial n-10 matches his goals be x

$$\text{so we get } \frac{(x+1)}{n} = 0.15$$

$$\text{we get } x+1 = 0.15n \quad (1)$$

From condition (2) we get :

$$\frac{(x+2)}{n} = 0.2$$

$$\text{we get } x+2 = 0.2n \quad (2)$$

Subtracting (1) and (2)

$$\text{we get } 1 = 0.05n$$

$$n = 20$$

So initially he played n-10 = 10 matches

 VIDEO SOLUTION

10. **92**

Let sum of marks of students be x
 Now therefore $x = 25 \times 50 = 1250$
 Now to maximize the marks of the toppers
 We will minimize the marks of 20 students
 so their scores will be (30, 31, 32, ..., 49)
 let score of toppers be y
 so we get $5y + \frac{20}{2}(79) = 1250$
 we get $5y + 790 = 1250$
 $5y = 460$
 $y = 92$
 So scores of toppers = 92

 VIDEO SOLUTION



11. B

Let us assume the family spends Rs. 100 each month for the first 3 months and then spends Rs. 50 in each of the next two months.

Then amount of onions bought = 10, 5, 4, 2, 1, for months 1-5 respectively.

Total amount bought = 22kg.

Total amount spent = $100 + 100 + 100 + 50 + 50 = 400$.

Average expense = $\frac{400}{22} = \text{Rs. } 18.18 \approx 18$

 VIDEO SOLUTION

12. D

The average marks for all the students is 38.

Sum = $5 \times 38 = 190$

To find the minimum marks scored by Amit, we need to maximise the score of remaining students.

Maximum scores sum of remaining students = $50 + 49 + 48 + 32 = 179$

Minimum possible score of Amit = $190 - 179 = 11$

It is given, Amit scored least. This implies maximum possible score of Amit is 31.

Difference = $31 - 11 = 20$

The answer is option D.

 VIDEO SOLUTION

13. C

It is given that the average of the 30 integers = 5

Sum of the 30 integers = $30 \times 5 = 150$

There are exactly 20 integers whose value is less than 5.

To maximise the average of the 20 integers, we have to assign minimum value to each of the remaining 10 integers

So the sum of 10 integers = $10 \times 6 = 60$

The sum of the 20 integers = $150 - 60 = 90$

Average of 20 integers = $\frac{90}{20} = 4.5$

 VIDEO SOLUTION

14. **B**

Total marks = $80 \times 10 = 800$

Total marks except highest and lowest marks = $81 \times 8 = 648$

So Summation of highest marks and lowest marks will be = $800 - 648 = 152$

When highest marks is 92, lowest marks will be = $152 - 92 = 60$

 VIDEO SOLUTION

15. **A**

Let the fixed income be x and the number of boarders be y .

$$x + 25y = 17500$$

$$x + 50y = 30000$$

$$\Rightarrow y = 500 \text{ and } x = 5000$$

$$x + 100y = 5000 + 50000 = 55000$$

$$\text{Average expense} = \frac{55000}{100} = \text{Rs. } 550.$$

 VIDEO SOLUTION



16. **B**

The first five numbers could be $n-2, n-1, n, n+1, n+2$. The next two number would then be, $n+3$ and $n+4$, in which case, the average of all the 7 numbers would be $\frac{(5n+2n+7)}{7} = n+1$

 VIDEO SOLUTION

17. **B**

Let x, y and z be no. of students in class X, Y, Z respectively.

From 1st condition we have

$$83x + 76y = 79x + 79y \text{ which give } 4x = 3y.$$

$$\text{Next we have } 76y + 85z = 81(y+z) \text{ which give } 4z = 5y.$$

Now overall average of all the classes can be given as $\frac{83x+76y+85z}{x+y+z}$

Substitute the relations in above equation we get,

$$\frac{83x+76y+85z}{x+y+z} = (83 \cdot \frac{3}{4} + 76 + 85 \cdot \frac{5}{4}) / (\frac{3}{4} + 1 + \frac{5}{4}) = 978/12 = 81.5$$

 VIDEO SOLUTION

18. **B**

Let the actual average be n . So, the new average is $n + 1.8$

Actual total = $10n$

New total = $10n + 18$

Let the number which was miswritten = ab (a is the tenth's digit and b is the units digit) = $10a+b$

and reversed number $ba = 10b+a$

So, $10b + a - (10a + b) = 18$

$\Rightarrow 9(b-a) = 18$

$\Rightarrow b-a = 2$

 VIDEO SOLUTION

19. E

Ten years ago, the total age of the family is 231 years.

Seven years ago, (Just before the death of the first person), the total age of the family would have been $231+8*3 = 231+24 = 255$.

This is because, in 3 years, every person in the family would have aged by 3 years,

Total change in age = $231+24 = 255$

After the death of one member, the total age is $255-60 = 195$ years.

Since a child takes birth in the same year, the number of members remain the same i.e. $(7+1) = 8$

Four years ago, (i.e. 6 years after start date) one of the member of age 60 dies, therefore, total age of the family is $195+24-60 = 159$ years.

Since a child takes birth in the same year, the number of members remain the same i.e. $(7+1) = 8$

After 4 more years, the current total age of the family is $= 8*4 + 159 = 191$ years

The average age is $191/8 = 23.875$ years ≈ 24 years (approx)

Alternatively,

Since the number of members is always the same throughout

The 2 older members dropped their age by 60

So, after 10yrs, total age $= 231 + 8*10 - 2*60 = 191$

Average age $= 191/8 = 23.875 \approx 24$

 VIDEO SOLUTION

20. B

The odd numbers in the set are 3, 5, 7, ..., $2n+1$

Sum of the odd numbers $= 3+5+7+\dots+(2n+1) = n^2 + 2n$

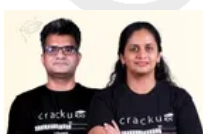
Average of odd numbers $= n^2 + 2n/n = n+2$

Sum of even numbers $= 2 + 4 + 6 + \dots + 2n = 2(1+2+3+\dots+n) = 2*n*(n+1)/2 = n(n+1)$

Average of even numbers $= n(n+1)/n = n+1$

So, difference between the averages of even and odd numbers $= 1$

 VIDEO SOLUTION



CAT Daily Articles



21. A

Let, the average score of boys in the mid semester exam is A .

Therefore, the average score of girls in the mid semester exam be $A+5$.

Hence, the total marks scored by the class is $20 \times (A) + 30 \times (A + 5) = 50 \times A + 150$

The average score of the entire class is $\frac{(50 \times A + 150)}{50} = A + 3$

wkt, class average increased by 2, class average in final term = $(A + 3) + 2 = A + 5$

Given, that score of girls dropped by 3, i.e $(A + 5) - 3 = A + 2$

Total score of girls in final term = $30 \times (A + 2) = 30A + 60$

	MID-TERM		FINAL-TERM	
No Boys/Girls	Avg score	Total score	Avg score	Total score
20	A	$20A$		
30	$A+5$	$30A+150$	$A+2$	$30A+60$
Class score		$50A+150$		
Class Average		$A+3$		$A+5$

Total class score in final term = $(A + 5) \times 50 = 50A + 250$

the total marks scored by the boys is $(50A + 250) - (30A - 60) = 20A + 190$

Hence, the average of the boys in the final exam is $\frac{(20G + 190)}{20} = A + 9.5$

	MID-TERM		FINAL-TERM	
No Boys/Girls	Avg score	Total score	Avg score	Total score
20	A	$20A$	$A+9.5$	$20A+190$
30	$A+5$	$30A+150$	$A+2$	$30A+60$
Class score		$50A+150$		$50A+250$
Class Average		$A+3$		$A+5$

Hence, the increase in the average marks of the boys is $(A + 9.5) - A = 9.5$

VIDEO SOLUTION

22. C

Let the average height of 22 toddlers be $3x$.

Sum of the height of 22 toddlers = $66x$

Hence average height of the two toddlers who left the group = x

Sum of the height of the remaining 20 toddlers = $66x - 2x = 64x$

Average height of the remaining 20 toddlers = $64x/20 = 3.2x$

Difference = $0.2x = 2$ inches $\Rightarrow x = 10$ inches

Hence average height of the remaining 20 toddlers = $3.2x = 32$ inches

VIDEO SOLUTION

23. 11

It is given that the maximum weight limit is 630. The lightest person's weight is 53 Kg and the heaviest person's weight is 57 Kg.

In order to have maximum people in the lift, all the remaining people should be of the lightest weight possible, which is 53 Kg.

Let there be n people.

$$53 + n(53) + 57 = 630$$

n is approximately equal to 9.8. Hence, 9 people are possible.

Therefore, a total of $9 + 2 = 11$ people can use the elevator.

 VIDEO SOLUTION

24. 60

Let the total number of tests be ' n ' and the average by ' A '

$$\text{Total score} = nA$$

When 1st 10 tests are excluded, decrease in total value of scores = $(nA - 20 * 10) = (nA - 200)$

$$\text{Also, } (n - 10)(A + 1) = (nA - 200)$$

On solving, we get $10A - n = 190$(i)

When last 10 tests are excluded, decrease in total value of scores = $(nA - 30 * 10) = (nA - 300)$

$$\text{Also, } (n - 10)(A - 1) = (nA - 300)$$

On solving, we get $10A + n = 310$(ii)

From (i) and (ii), we get $n = 60$

Hence, 60 is the correct answer.

 VIDEO SOLUTION

25. D

The possible average age of people whose ages are below 51 years will be maximum if the average age of the number of people aged 51 years and above is minimum. Hence, we can say that there are 30 people having same age 51 years.

Let ' x ' be the maximum average age of people whose ages are below 51.

Then we can say that,

$$\frac{51 * 30 + 39 * x}{30 + 39} = 38$$

$$\Rightarrow 1530 + 39x = 2622$$

$$\Rightarrow x = 1092/39 = 28$$

Hence, we can say that option D is the correct answer.

 VIDEO SOLUTION

100+ CAT Quant Questions



26. B

Assume the average of 21 students other than Ramesh = a

Sum of the scores of 21 students other than Ramesh = $21a$

Hence the average of 22 students = $a+1$

Sum of the scores of all 22 students = $22(a+1)$

The score of Ramesh = Sum of scores of all 22 students - Sum of the scores of 21 students other than Ramesh
 $= 22(a+1) - 21a = a+22 = 82.5$ (Given)

$$\Rightarrow a = 60.5$$

Hence, sum of the scores of all 22 students = $22(a+1) = 22 \times 61.5 = 1353$

Now the sum of the scores of students other than Gautam = $21 \times 62 = 1302$

Hence the score of Gautam = $1353 - 1302 = 51$

 VIDEO SOLUTION

27. 15

Let us assume the four numbers to be a, b, c and d in ascending order.

Average of first two numbers is 1 more than the first number

$$\frac{(a+b)}{2} = a + 1$$

$$b - a = 2$$

$$b = a + 2$$

Average of first three numbers is 2 more than average of first two numbers

$$\frac{(a+b+c)}{3} = \frac{(a+b)}{2} + 2$$

$$2c = a + b + 12$$

Substituting the value for b

$$2c = a + a + 2 + 12$$

$$2c = 2a + 14$$

$$c = a + 7$$

Average of first four numbers is 3 more than average of first three numbers.

$$\frac{(a+b+c+d)}{4} = \frac{(a+b+c)}{3} + 3$$

$$3d = a + b + c + 36$$

Substituting the value of b and c

$$3d = a + a + 2 + a + 7 + 36$$

$$3d = 3a + 45$$

$$d = a + 15$$

d is the largest and a is the smallest and we know that $d = a + 15$

Hence the difference between the smallest and the largest values is 15.

 VIDEO SOLUTION

28. 70

We are given that average of three distinct integers is 28, that means the sum of these three integers is $28 \times 3 = 84$

Let us write $x + y + z = 84$

x, y, z being the three distinct integers in ascending order.

If the smallest number is increased by 7 and the largest number is reduced by 10

$$(x + 7) + (y) + (z - 10) = 81$$

New arithmetic mean will be $\frac{81}{3} = 27$

And this is said to be 2 more than the middle number, meaning

$$27 - 2 = y = 25$$

$$x + z = 59$$

We are given that difference between the largest and the smallest numbers becomes 64,

$$(z - 10) - (x + 7) = 64$$

$$z - x = 81$$

Adding the two equations we get, $2z = 140$

$$z = 70$$

 VIDEO SOLUTION

29. C

Let the average salary of managers be x and let the average salary of engineers be y . The total salary of all the employees will be $(5 + 25) * 60000 = 1800000$

We have, $5x + 25y = 1800000$ (1)

If the average salary of all the employees increases by 5%, the total salary of all the employees will also increase by 5%, because the total number of employees remains the same. The new total salary will be, $1800000 \times 1.05 = 1890000$

Also, the average salary of all the managers has increased by 20% and has become $1.2x$, we have

$$5 * 1.2x + 25y = 1890000, \text{ or } 6x + 25y = 1890000 \text{ (2)}$$

Subtracting equation (1) from equation (2), we get,

$$x = 90000$$

$$\text{Which gives } 25y = 1800000 - 90000 * 5 \text{ or } y = \frac{1350000}{25} = 54000$$

Therefore, the correct answer is option C.

 VIDEO SOLUTION

30. 49

Let the copies sold on days 1 through 9 be $d_1, \dots, d_9$

From the averages:

$$d_1 + \dots + d_7 = 7 \times 60 = 420$$

$$d_1 + \dots + d_8 = 8 \times 63 = 504$$

$$\text{So } d_8 = 504 - 420 = 84. \text{ Then } d_9 = 84 - 11 = 73$$

The sum from day 2 to day 9 is $8 \times 66 = 528$

$$d_2 + \dots + d_9 = (d_1 + \dots + d_8) - d_1 + d_9 = 504 - d_1 + 73 = 577 - d_1$$

$$\text{Thus } 577 - d_1 = 528 \Rightarrow d_1 = 49$$

 VIDEO SOLUTION

 **100+ CAT DILR Questions**

